

Mizer model

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Loading library

Load the packages required to design mizer model and produce figures.

```
library(mizer)
library(mizerExperimental)
library(tidyverse)
```

Setting initial parameters

```
params_single <- newSingleSpeciesParams()

params_initial <- setResource(params_single,
                              resource_dynamics = "resource_semichemostat")

params_double <- params_initial
params_double <- setResource(params_double,
                              resource_level = 0.1)

initialN(params_double) <- 2*initialN(params_double)
initialNResource(params_double) <- initialNResource(params_double)/2
```

Implementing fishing gear

```
gear_params(params_double) <- data.frame(
  gear = "gear",
  species = "Target species",
  catchability = 0.3,
```

```

sel_func = "sigmoid_weight",
sigmoidal_weight = 15,
sigmoidal_sigma = 5)

params_double <- setFishing(params_double, gear_params = gear_params)

gear_params(params_double)

```

	gear	species	catchability	sel_func
Target species, gear	gear	Target species	0.3	sigmoid_weight
		sigmoidal_weight	sigmoidal_sigma	
Target species, gear		15	5	

Simulating ecosystem behaviour

Flux of system

```

sim_double <- project(params_double, t_max = 15, effort = 1)

```

Calculate flux

```

N <- finalN(sim_double)["Target species", , drop = TRUE]
w <- w(params_double)

E_growth <- getEGrowth(params_double)["Target species", , drop = TRUE]
gr <- w * E_growth
flux <- gr * N

```

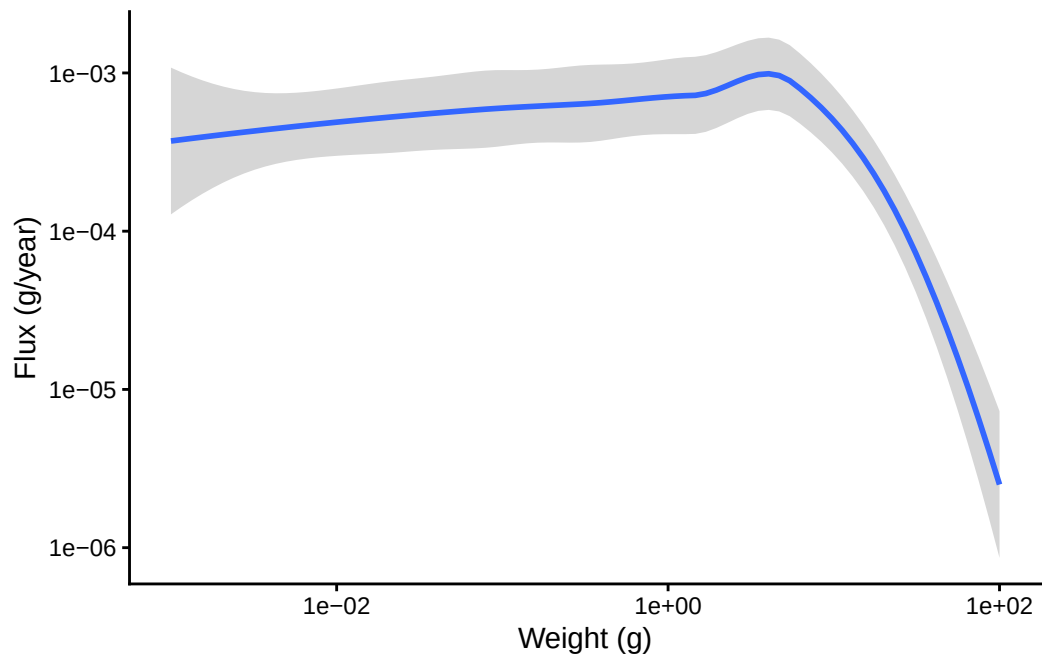
Plot flux

```

initial_flux_data <- data.frame(Weight = w,
                                Flux = flux)
initial_flux_plot <- ggplot(initial_flux_data,
                             aes(x = Weight, y = Flux)) +
  geom_smooth() +
  scale_x_log10() +
  scale_y_log10() +

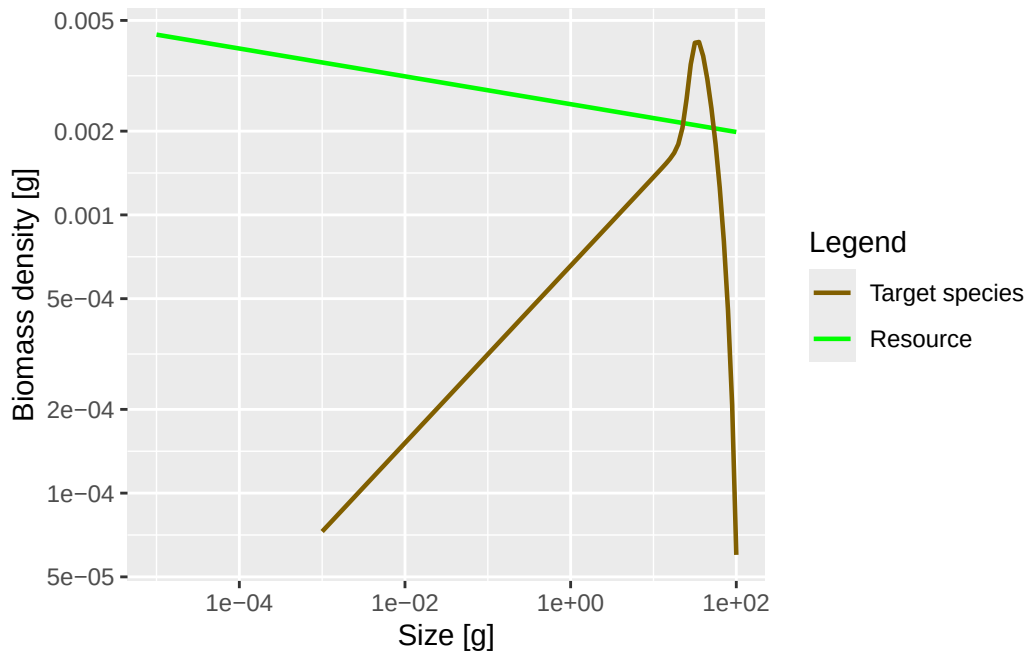
```

```
labs(
  x = paste0("Weight (g)"),
  y = paste0("Flux (g/year)")) +
theme_classic()
initial_flux_plot
```



Biomass density

```
plotSpectra(params_double, power = 2)
```



Locally reduce parameters

Reduce resource rate

```
rr <- resource_rate(params_double)
w_full <- w_full(params_double)
w_full[222:242]
```

```
[1] 0.1000000 0.1122018 0.1258925 0.1412538 0.1584893 0.1778279 0.1995262
[8] 0.2238721 0.2511886 0.2818383 0.3162278 0.3548134 0.3981072 0.4466836
[15] 0.5011872 0.5623413 0.6309573 0.7079458 0.7943282 0.8912509 1.0000000
```

```
rr[222:242] <- rr[222:242] / 100000000

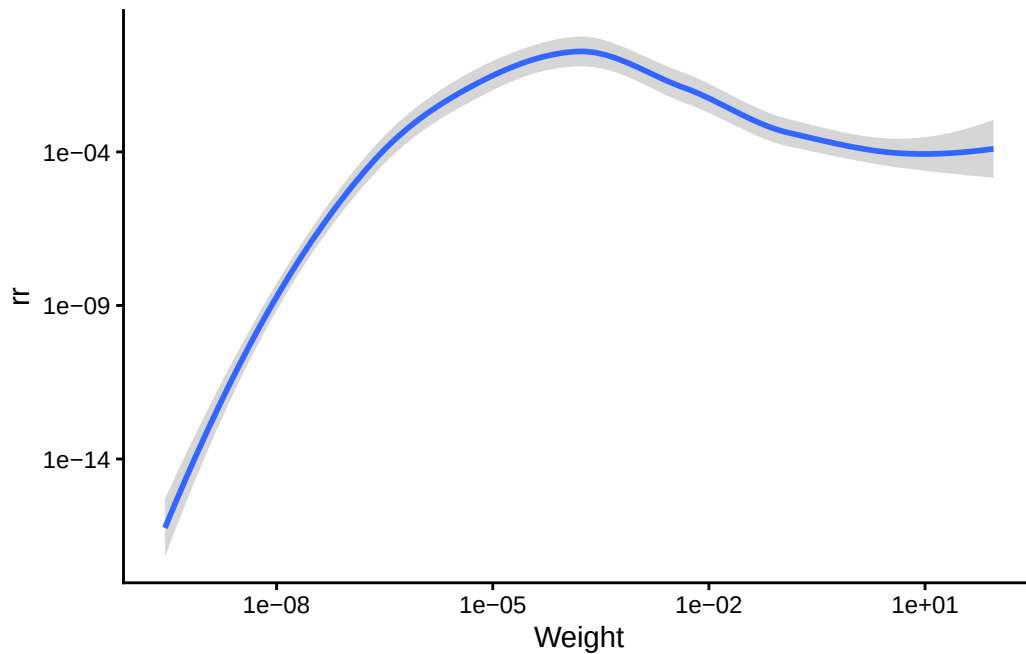
rr_data <- data.frame(
  Weight = params_double@w_full,
  rr = rr)

rr_plot <- ggplot(rr_data,
  aes(x = Weight, y = rr)) +
  geom_smooth() +
```

```

scale_x_log10() +
scale_y_log10() +
theme_classic()
rr_plot

```



Reduce resource capacity

```

rc <- resource_capacity(params_double)
w_full[222:242]

```

```

[1] 0.1000000 0.1122018 0.1258925 0.1412538 0.1584893 0.1778279 0.1995262
[8] 0.2238721 0.2511886 0.2818383 0.3162278 0.3548134 0.3981072 0.4466836
[15] 0.5011872 0.5623413 0.6309573 0.7079458 0.7943282 0.8912509 1.0000000

```

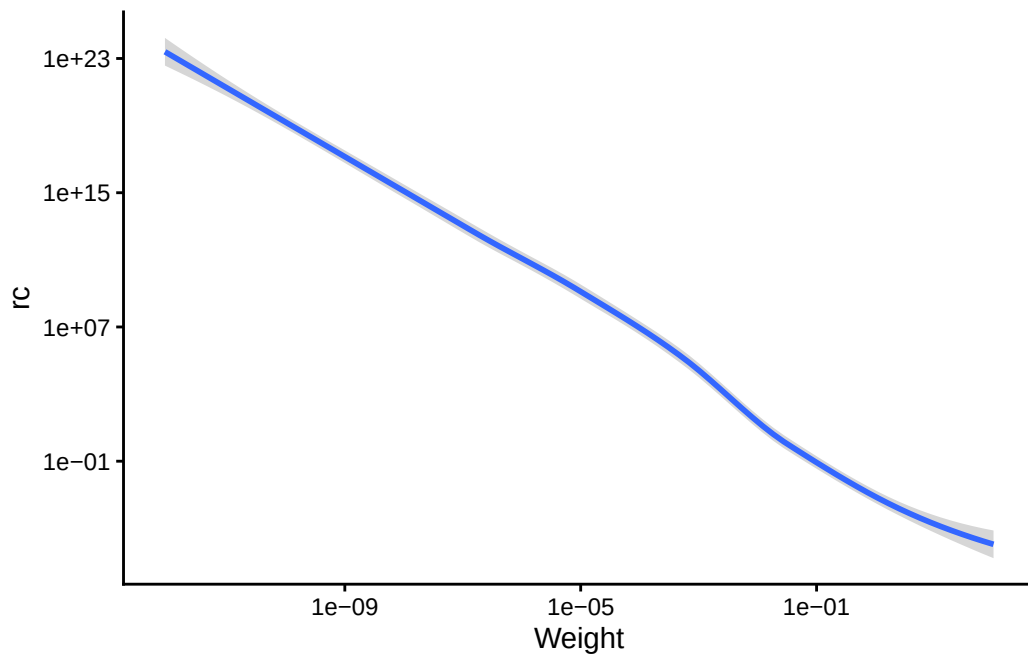
```

rc[222:242] <- rc[222:242] / 100000000

rc_data <- data.frame(
  Weight = params_double@w_full,
  rc = rc)

```

```
rc_plot <- ggplot(rc_data,
                  aes(x = Weight, y = rc)) +
  geom_smooth() +
  scale_x_log10() +
  scale_y_log10() +
  theme_classic()
rc_plot
```



Set new parameters with local resource parameter reductions

```
params_reduced <- setResource(params_double,
                              resource_capacity = rc,
                              resource_rate = rr,
                              balance = FALSE)
```

Predation kernel

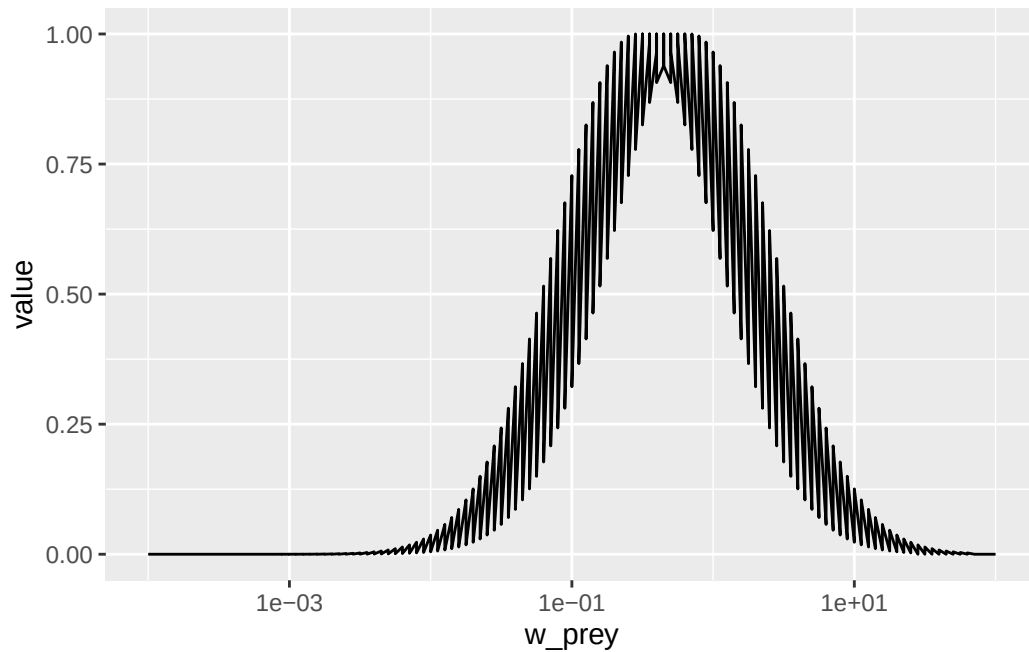
Define predation kernel

```

pred_kernel <- getPredKernel(params_reduced)
pred_kernel_reduced <- pred_kernel[, 90:98, , drop = FALSE]

ggplot(melt(pred_kernel_reduced)) +
  geom_line(aes(x = w_prej, y = value)) +
  scale_x_log10(limits = c(1e-4, 100))

```



Narrow predation kernel

```
select(species_params(params_reduced), beta, sigma)
```

```

      beta sigma
Target species 100  1.3

```

```

params <- params_reduced

given_species_params(params)$sigma <- 0.2
given_species_params(params)$beta <- 100

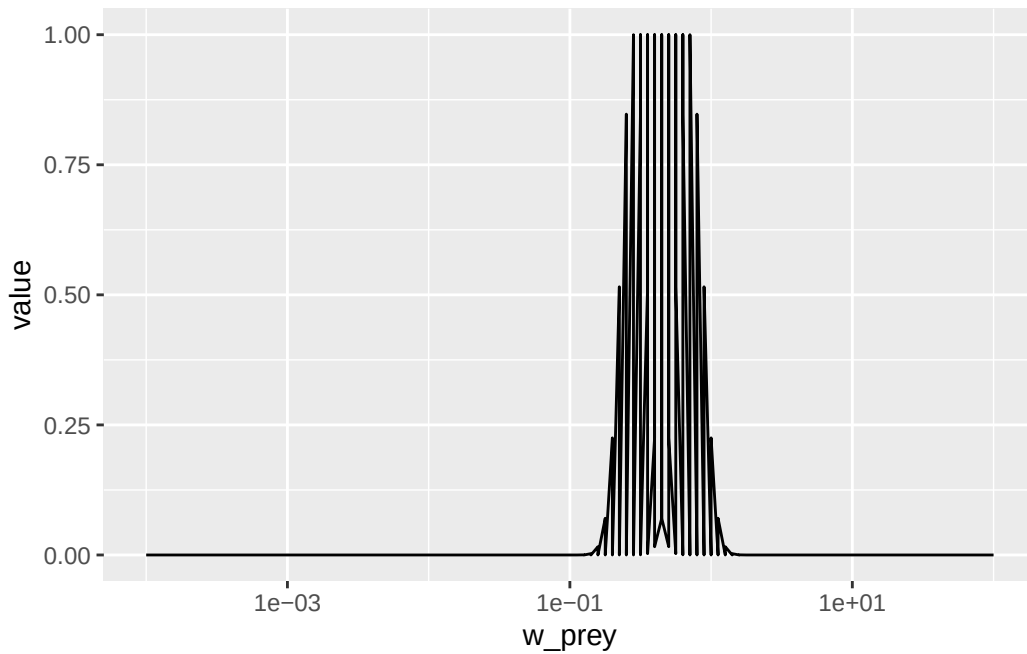
getPredKernel(params)[, 90:98, , drop = FALSE] %>%

```

```

melt() %>%
  ggplot() +
  geom_line(aes(x = w_pre, y = value)) +
  scale_x_log10(limits = c(1e-4, 100))

```



New flux of system

```

sim_reduced <- project(params, t_max = 15, effort = 1)

N_reduced <- finalN(sim_reduced)["Target species", , drop = TRUE]
w <- w(params_reduced)

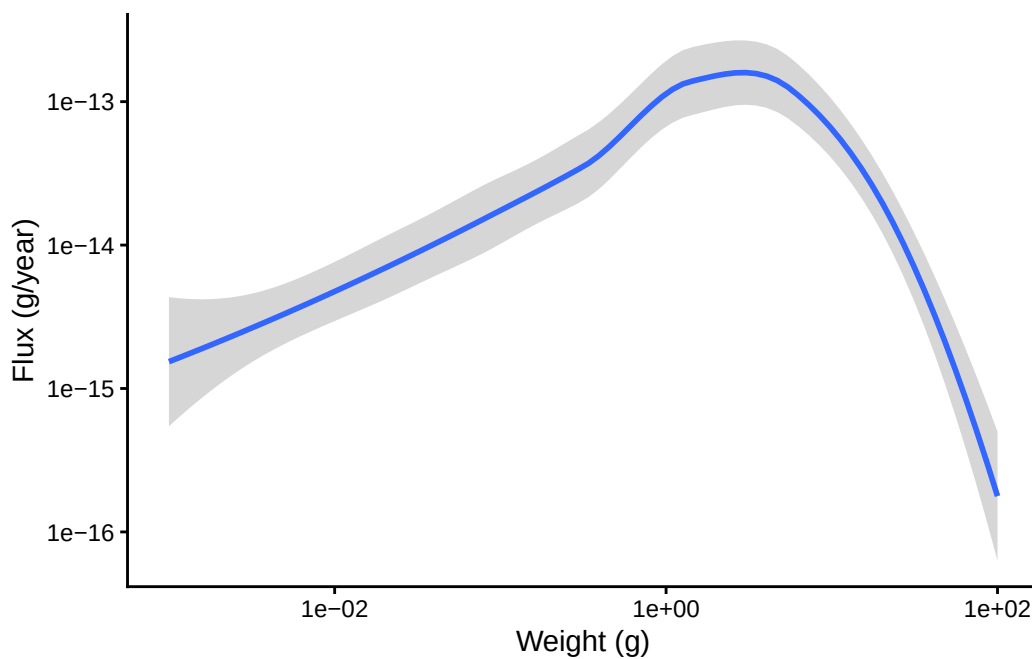
E_growth_reduced <- getEGrowth(params)["Target species", , drop = TRUE]
grr <- w * E_growth_reduced
flux_reduced <- grr * N_reduced

reduced_flux_data <- data.frame(Weight = w,
                                Flux = flux_reduced)

```


Plot flux on a log-log axis

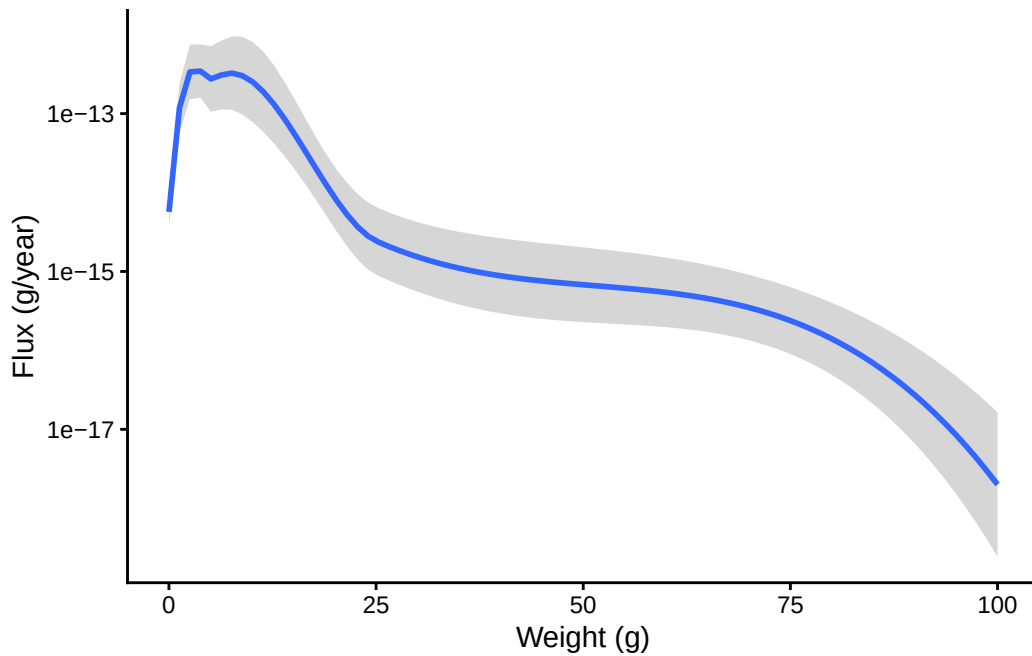
```
reduced_flux_log_plot <- ggplot(reduced_flux_data,  
                                aes(x = Weight, y = Flux)) +  
  geom_smooth() +  
  scale_y_log10() +  
  scale_x_log10() +  
  labs(  
    x = paste0("Weight (g)"),  
    y = paste0("Flux (g/year)")) +  
  theme_classic()  
reduced_flux_log_plot
```



Plot flux on a log y-axis

```
reduced_flux_plot <- ggplot(reduced_flux_data,  
                             aes(x = Weight, y = Flux)) +  
  geom_smooth() +  
  scale_y_log10() +  
  labs(  
    x = paste0("Weight (g)"),  
    y = paste0("Flux (g/year)")) +
```

```
theme_classic()
reduced_flux_plot
```

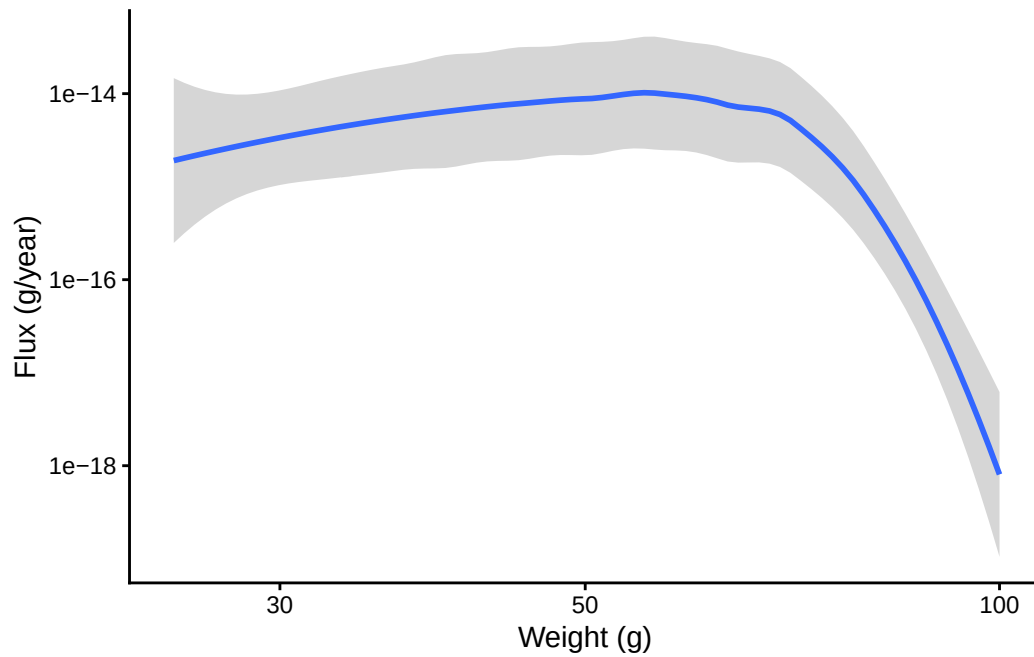


Flux over a limited weight range

```
limited_flux_data <- data.frame(Weight = w[89:101],
                               Flux = flux_reduced[89:101])
```

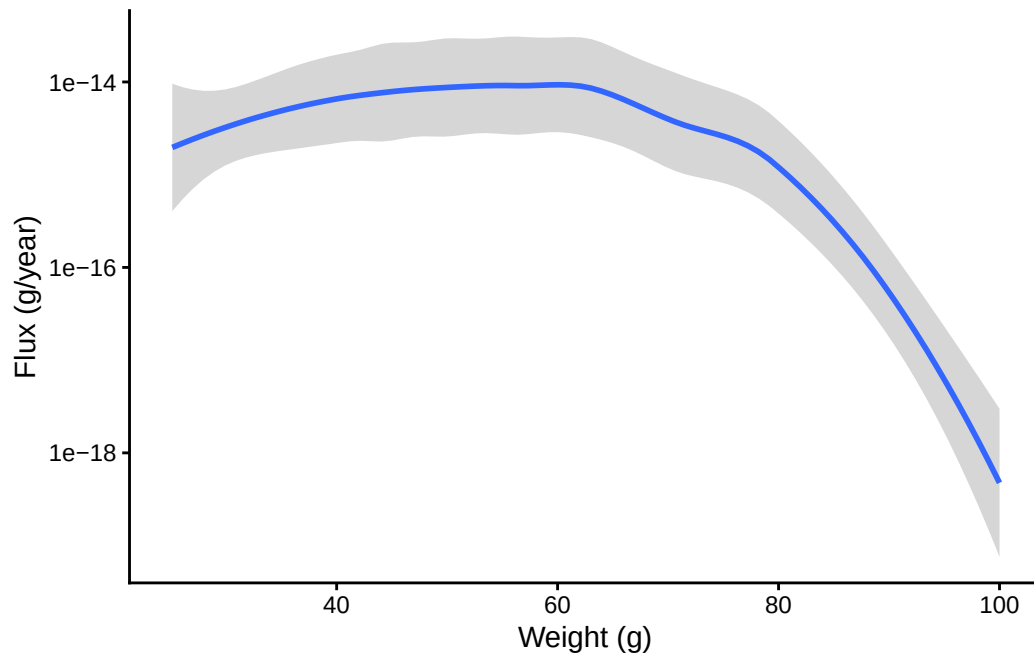
Plot on a log-log axis

```
limited_flux_log_plot <- ggplot(limited_flux_data,
                               aes(x = Weight, y = Flux)) +
  geom_smooth() +
  scale_y_log10() +
  scale_x_log10(limits = c(25, 100)) +
  labs(
    x = paste0("Weight (g)"),
    y = paste0("Flux (g/year)")) +
  theme_classic()
limited_flux_log_plot
```



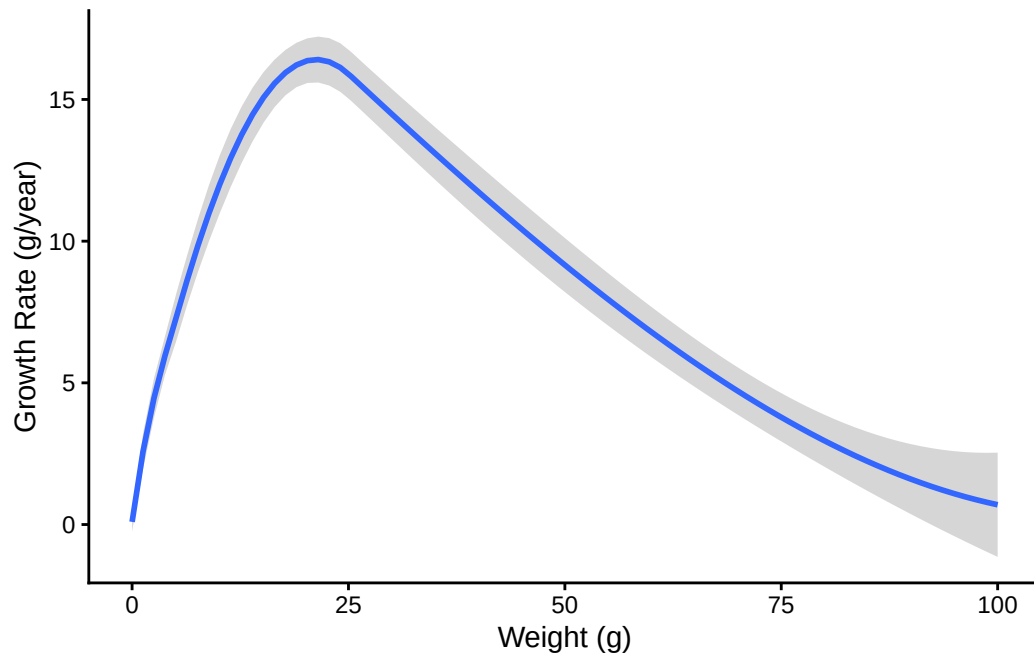
Plot on a log y-axis

```
limited_flux_plot <- ggplot(limited_flux_data,  
                           aes(x = Weight, y = Flux)) +  
  geom_smooth() +  
  scale_y_log10() +  
  scale_x_continuous(limits = c(25, 100)) +  
  labs(  
    x = paste0("Weight (g)"),  
    y = paste0("Flux (g/year)")) +  
  theme_classic()  
limited_flux_plot
```



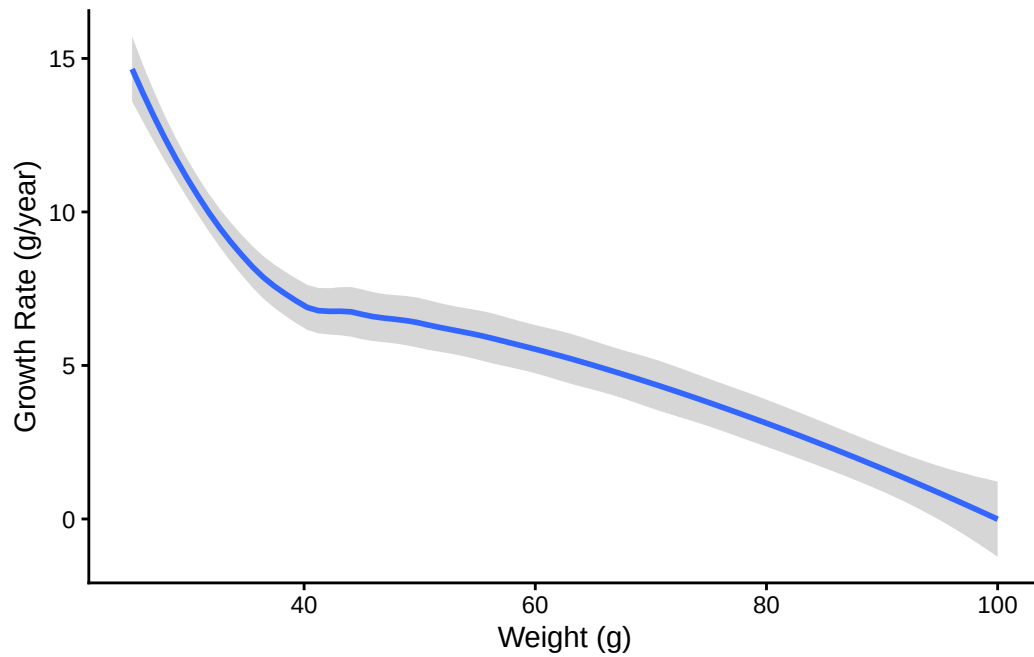
Plot growth rate

```
growth_data <- data.frame(growth = E_growth_reduced,
                          weight = w)
growth_plot <- ggplot(growth_data,
  aes(x = weight,
      y = growth)) +
  geom_smooth() +
  labs(x = paste0("Weight (g)"),
      y = paste0("Growth Rate (g/year)")) +
  theme_classic()
growth_plot
```



Plot growth rate over a limited weight range

```
limited_growth_data <- data.frame(growth = E_growth_reduced[89:101],  
                                weight = w[89:101])  
limited_growth_plot <- ggplot(limited_growth_data,  
                             aes(x = weight,  
                                 y = growth)) +  
  geom_smooth() +  
  labs(x = paste0("Weight (g)"),  
       y = paste0("Growth Rate (g/year)")) +  
  theme_classic()  
limited_growth_plot
```



Plot yield

```
yield <- getYield(sim_reduced)
time <- getTimes(sim_reduced)

yield_data <- data.frame(Time = time,
                          Yield = yield)

yield_graph <- ggplot(yield_data,
                      aes(x = Time, y = Target.species)) +
  geom_line() +
  labs(x = paste0("Year"),
       y = paste0("Yield (g/year)") ) +
  theme_classic()
yield_graph
```

